WORKED SOLUTION

STEP III 1994, Question 8

Pure • Four-Square Identity

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 7.8, *Four Squares*: https://beyondstep.keithmcnulty.org/ch07/four-squares.html

The key idea

A matrix of the form $\begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix}$ has determinant $zz^* + ww^* = |z|^2 + |w|^2$, which is a sum of *four* real squares. If we pack a, b, c, d into one such matrix and p, q, r, s into another, then $\det(AB) = \det A \det B$ gives the left-hand side. The work is to show that the product AB has the same special form, because then its determinant is also a sum of four squares, and those squares are the L_i^2.

Solution

Setting up

For complex numbers z and w write

$$M(z,w) = \begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix}, \qquad \det M(z,w) = zz^* - w(-w^*) = |z|^2 + |w|^2.$$

Choose

$$z_1 = a + b\mathrm{i}, \quad z_2 = c + d\mathrm{i}, \quad z_3 = p + q\mathrm{i}, \quad z_4 = r + s\mathrm{i}.$$

Then

$$\det M(z_1, z_2) = a^2 + b^2 + c^2 + d^2, \qquad \det M(z_3, z_4) = p^2 + q^2 + r^2 + s^2.$$

The product has the same form

Multiplying out,

$$M(z_1, z_2)\, M(z_3, z_4) = \begin{pmatrix} z_1 z_3 - z_2 z_4^* & z_1 z_4 + z_2 z_3^* \\ -z_2^* z_3 - z_1^* z_4^* & -z_2^* z_4 + z_1^* z_3^* \end{pmatrix}.$$

Put $w_1 = z_1 z_3 - z_2 z_4^*$ and $w_2 = z_1 z_4 + z_2 z_3^*$, the top row. Since the conjugate of a product is the product of the conjugates, and $(z^*)^* = z$,

$$w_1^* = z_1^* z_3^* - z_2^* z_4, \qquad -w_2^* = -z_1^* z_4^* - z_2^* z_3,$$

and these are exactly the bottom-right and bottom-left entries. So

$$M(z_1, z_2)\, M(z_3, z_4) = M(w_1, w_2).$$

Ask yourself Why does it matter that the product has this special shape?

For a general 2×2 complex matrix the determinant is not a sum of squares at all. It is only because the product is again of the form $M(w_1, w_2)$ that its determinant is $|w_1|^2 + |w_2|^2$, which splits into four real squares.

Taking determinants

Using $\det(AB) = \det A \, \det B$,

$$(a^2 + b^2 + c^2 + d^2)(p^2 + q^2 + r^2 + s^2) = \det M(w_1, w_2) = |w_1|^2 + |w_2|^2.$$

It remains to find the real and imaginary parts of w_1 and w_2:

$$\begin{aligned} w_1 &= (a + b\mathrm{i})(p + q\mathrm{i}) - (c + d\mathrm{i})(r - s\mathrm{i}) \\ &= \big[(ap - bq) + (aq + bp)\mathrm{i}\big] - \big[(cr + ds) + (dr - cs)\mathrm{i}\big] \\ &= (ap - bq - cr - ds) + (aq + bp + cs - dr)\mathrm{i}, \end{aligned}$$

$$\begin{aligned} w_2 &= (a + b\mathrm{i})(r + s\mathrm{i}) + (c + d\mathrm{i})(p - q\mathrm{i}) \\ &= \big[(ar - bs) + (as + br)\mathrm{i}\big] + \big[(cp + dq) + (dp - cq)\mathrm{i}\big] \\ &= (ar - bs + cp + dq) + (as + br - cq + dp)\mathrm{i}. \end{aligned}$$

So we may take

$$L_1 = ap - bq - cr - ds, \qquad L_2 = aq + bp + cs - dr,$$
$$L_3 = ar - bs + cp + dq, \qquad L_4 = as + br - cq + dp,$$

and then $|w_1|^2 + |w_2|^2 = L_1^2 + L_2^2 + L_3^2 + L_4^2$, giving

$$(a^2 + b^2 + c^2 + d^2)(p^2 + q^2 + r^2 + s^2) = L_1^2 + L_2^2 + L_3^2 + L_4^2.$$

Each L_i is a sum of terms (one letter from a, b, c, d) $\times$ (one letter from p, q, r, s), so it is linear in each set of variables separately, as required. (The answer is not unique: changing any signs of the L_i, or choosing the z_j differently, gives other valid sets.)

Checks

- Take $(a,b,c,d) = (1,2,3,4)$ and $(p,q,r,s) = (5,6,7,8)$. The left-hand side is $30 \times 174 = 5220$. The formulae give $(L_1, L_2, L_3, L_4) = (-60, 12, 30, 24)$, and $3600 + 144 + 900 + 576 = 5220$.
- Put $c = d = r = s = 0$. Then $L_3 = L_4 = 0$ and the identity reduces to $(a^2+b^2)(p^2+q^2) = (ap-bq)^2 + (aq+bp)^2$, the two-square identity from $|z_1 z_3| = |z_1||z_3|$.
- A computer check confirmed the identity exactly for 20,000 random rational choices of the eight variables, and confirmed $M(z_1,z_2)M(z_3,z_4) = M(w_1,w_2)$ numerically.

Common slips

- Forgetting the conjugates in the bottom row when multiplying, which breaks the special form of the product.
- Sign errors in $(c+d\mathrm{i})(r-s\mathrm{i})$: the term $-ds\,\mathrm{i}^2$ becomes $+ds$.

Beyond STEP

This is Euler's four-square identity. The matrices $M(z,w)$ are a concrete model of Hamilton's quaternions: they are closed under multiplication, and the determinant is the squared "length", which is multiplicative. The identity is the key step in Lagrange's theorem that every positive integer is a sum of four squares, since it reduces the problem to primes. Davenport's *The Higher Arithmetic* covers sums of squares, and Needham's *Visual Complex Analysis* gives geometric intuition for the two-square case.

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